

ADAPT-Rehab: A Multimodal AI System for Real-Time Elderly Rehabilitation with 3D Pose Estimation and Voice-Interactive LLM Coaching

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Abstract—Vietnam’s elderly population (aged 60+) is projected to reach 20% by 2030, yet the physiotherapist-to-population ratio remains as low as 0.4 per 100,000, with a severe urban-rural disparity. We present ADAPT-REHAB, a multimodal AI rehabilitation system designed for elderly Vietnamese users. The system integrates: (1) real-time whole-body 3D pose estimation via RTMW3D-L [1] from MMPose [2], producing 133 keypoints (body, hands, and face in COCO-WholeBody format) at 129.7 FPS with quaternion-based kinematics for multi-plane joint angle computation; (2) multi-indicator facial state detection using OpenFace 3.0 with Action Unit analysis, PSPI-based pain intensity estimation [3], PERCLOS-based fatigue monitoring [4], and engagement detection [5]; (3) voice-interactive coaching powered by Large Language Models with Retrieval-Augmented Generation for personalized, clinically-grounded exercise guidance in Vietnamese; and (4) a six-dimension scoring framework incorporating SPARC smoothness metrics [6], constrained DTW alignment, and temporal compensation detection. The system achieves 129.7 FPS on GPU hardware with whole-body pose coverage, and includes a comprehensive evaluation framework with proper MPJPE protocols, cross-skeleton mapping for benchmark compatibility, and ICC-based reliability analysis. ADAPT-REHAB is the first system to combine whole-body 3D pose estimation, Vietnamese voice interaction, multi-indicator facial state detection, and LLM-based coaching for elderly rehabilitation.

Index Terms—rehabilitation, pose estimation, elderly care, multimodal AI, LLM coaching, Vietnamese, pain detection, whole-body tracking, RTMW3D, SPARC

I. INTRODUCTION

Vietnam is experiencing rapid demographic aging, with the elderly population (aged 60+) projected to reach 20% by 2030, classifying the country as an “aged society” [7]. This demographic shift, occurring before economic affluence is achieved, creates unprecedented challenges for healthcare infrastructure, particularly in physical rehabilitation. The current physiotherapist-to-population ratio stands at approx-

imately 0.4–0.5 per 100,000, with a severe 5–10 $\times$ disparity between urban and rural areas [8].

Computer vision-based rehabilitation systems have shown promise in augmenting physiotherapy services [9], [10]. However, existing systems face several limitations when applied to elderly Vietnamese users: (1) reliance on expensive depth sensors (Kinect, LiDAR) that are inaccessible in home settings; (2) use of body-only 2D pose estimation that misses hand gestures and facial expressions; (3) absence of Vietnamese-language interfaces and voice guidance; (4) lack of elderly-specific adaptations for reduced mobility and pain sensitivity; and (5) generic feedback mechanisms that do not account for individual clinical conditions.

We present ADAPT-REHAB, a multimodal AI rehabilitation system that addresses these limitations through five key contributions:

- 1) **First Vietnamese-language elderly rehabilitation system:** Culturally adapted interface with Vietnamese voice instructions using respectful “Bac” (uncle/auntie) address forms, specifically designed for users aged 60+ with varying literacy levels.
- 2) **Real-time whole-body 3D pose estimation:** Integration of RTMW3D-L [1] from MMPose [2] for direct image-to-3D pose estimation producing 133 keypoints in COCO-WholeBody format: body (17 joints), foot (6 keypoints), face (68 landmarks in dlib convention), and hands (42 keypoints), achieving 129.7 FPS with quaternion-based joint angle computation for multi-plane accuracy.
- 3) **Multi-indicator facial state detection:** A comprehensive face analysis pipeline integrating OpenFace 3.0 for Action Unit detection, PSPI-based pain intensity estimation [3], PERCLOS-based fatigue monitoring [4], blink and yawn detection, and engagement analysis [5], with automatic fallback to geometric anal-

ysis.

- 4) **Voice-interactive LLM coaching:** Integration of Whisper ASR [11] and Edge-TTS with Large Language Models for natural language exercise guidance in Vietnamese, grounded in clinical knowledge via RAG for safety.
- 5) **Clinically-informed scoring framework:** Six-dimension scoring incorporating SPARC smoothness [6], quaternion kinematics, constrained DTW alignment, and temporal compensation detection, with a comprehensive evaluation framework supporting cross-skeleton benchmark compatibility via MPJPE protocols and ICC-based reliability analysis.

The remainder of this paper is organized as follows: Section II reviews related work. Section III details our system architecture and algorithms. Section IV describes the experimental setup and evaluation. Section V discusses implications and limitations. Section VI concludes with future directions.

II. RELATED WORK

A. Computer Vision-Based Rehabilitation

Automated rehabilitation systems have evolved from sensor-based to vision-based approaches. Commercial platforms such as Sword Health and Hinge Health use wearable sensors with clinical validation [12], while academic systems like Pose Trainer [9] demonstrated the feasibility of webcam-based exercise correction. Recent work on AIGC-based fitness feedback [10] integrated pose estimation with LLM-generated feedback, but focused on general fitness rather than clinical rehabilitation. Comprehensive surveys [13] highlight the growing trend of combining vision-based pose analysis with natural language feedback, though none specifically target elderly Vietnamese users.

B. 3D Pose Estimation for Rehabilitation

Building on these vision-based approaches, 3D pose estimation has become increasingly important for accurate joint angle measurement. MediaPipe BlazePose [14] remains widely used for rehabilitation due to its speed (300+ FPS), but produces non-metric z-coordinates with approximately 63mm MPJPE on Human3.6M [15]. State-of-the-art methods such as MotionBERT [16], MotionAGFormer [17], and MeTRAbs [18] achieve 35–39mm MPJPE, representing a 40–50% improvement. Recent work on whole-body pose estimation, particularly RTMW3D [1] from MMPose [2], extends this to 133 keypoints covering body, hands, and face in real-time, enabling comprehensive monitoring of exercise form including hand placement and facial expressions.

C. Pain and Emotion Detection

Beyond body pose, facial expression analysis provides critical information about user comfort during exercise. Facial Action Unit (AU) detection forms the basis for pain estimation through the Prkachin-Solomon Pain Intensity

(PSPI) index [3]. Deep learning approaches including JAA-Net [19] and ME-GraphAU [20] have improved AU detection accuracy on standard benchmarks. For emotion recognition, MobileNetV3-Large and ViT-based models achieve 87–93% on RAF-DB [21]. However, elderly faces present unique challenges: reduced expressiveness, skin texture confusion with wrinkles, and medication-induced facial masking [22], necessitating lightweight, interpretable approaches for real-time deployment.

Fatigue detection has been extensively studied in driving safety, with PERCLOS [4] (percentage of eye closure) being the most validated metric. The Eye Aspect Ratio (EAR) [23] provides a computationally efficient proxy for eye closure detection. Engagement detection using facial AU analysis has been proposed by Whitehill et al. (2014) [5] and D'Mello & Graesser (2010) [24].

D. LLM-Based Healthcare Assistance

Complementing visual analysis, Large Language Models have been explored for rehabilitation guidance [25]. RAG-grounded LLMs reduce hallucinations by anchoring responses in verified clinical literature [26]. However, current systems lack integration with real-time pose analysis and voice interaction, particularly for elderly users who benefit from auditory guidance [27]. The safety of LLM-based medical advice remains a concern [28], necessitating robust guardrails and contraindication checking.

E. Biomechanics Metrics for Rehabilitation

The quantitative assessment of exercise quality relies on biomechanics metrics. Joint angle computation methods range from simple dot-product (sagittal plane only) to quaternion-based approaches that handle multi-plane rotations without gimbal lock [29]. The International Society of Biomechanics (ISB) provides standardized joint coordinate systems [30] for clinical angle reporting. Movement smoothness metrics include jerk-based measures [31] and the Spectral Arc Length (SPARC), which is duration-independent and clinically validated for stroke rehabilitation [6]. Dynamic Time Warping (DTW) variants, including weighted and constrained DTW [32], [33], enable robust motion comparison across different execution speeds, which is essential for elderly users who may perform exercises at varying tempos.

III. METHODOLOGY

A. System Architecture

ADAPT-REHAB follows a five-layer pipeline architecture as illustrated in Fig. 1. The design prioritizes modularity, whole-body coverage, and real-time performance on consumer hardware.

- 1) **Input Layer:** Webcam capture (30 FPS), microphone for voice commands (Whisper ASR), and user profile (JSON with age, ROM limits, medical conditions).
- 2) **Perception Layer:** Whole-body 3D pose estimation via RTMW3D-L [1] (133 keypoints: body + hands +

face), face analysis via OpenFace 3.0 with Action Unit detection, and multi-indicator facial state detection.

- 3) **Analysis Layer:** Quaternion-based joint angle computation, SPARC smoothness measurement, weighted DTW alignment, temporal compensation detection, and multi-indicator fatigue analysis.
- 4) **Intelligence Layer:** LLM coaching with RAG grounding, safety guardrails, Vietnamese voice synthesis (Edge-TTS), and personalized feedback generation.
- 5) **Output Layer:** Visual overlays (skeleton with angle arcs), Vietnamese voice feedback, real-time metrics dashboard, and session reports.

The architecture supports graceful degradation: if the LLM API is unreachable, the coaching module uses rule-based Vietnamese feedback; the face analysis pipeline automatically falls back from OpenFace 3.0 to geometric analysis based on available dependencies.

B. Whole-Body 3D Pose Estimation

We employ RTMW3D-L [1] from MMPose [2] for real-time whole-body 3D pose estimation. RTMW3D-L is a top-down architecture based on a Cascade Top-Down framework with a CSPNeXt backbone, CSPNeXtPAFPN neck, and RTMWHead that predicts both 2D heatmaps and 3D relative coordinates.

The model produces 133 keypoints directly from RGB images:

$$\mathbf{P}_{3D} = f_{\text{RTMW3D-L}}(\mathbf{I}) \in \mathbb{R}^{133 \times 3} \quad (1)$$

where the output follows COCO-WholeBody format: body joints (indices 0–16, 17 points), foot keypoints (17–22, 6 points), face landmarks (23–90, 68 points in dlib convention), left hand (91–111, 21 points), and right hand (112–132, 21 points).

The model is pre-trained on the COCO-WholeBody dataset and fine-tuned on the Cocktail-14 multi-person 3D dataset. For inference, we use the MMPose `inference_topdown` API with PyTorch backend, achieving 129.7 FPS on an NVIDIA RTX 5880 Ada Generation GPU.

For joint angle computation, we use the body keypoints (indices 0–16) in COCO format. Given three consecutive joints $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ forming an angle at $\mathbf{b}$, the dot-product angle is:

$$\theta_{\text{dot}} = \arccos \left(\frac{(\mathbf{a} - \mathbf{b}) \cdot (\mathbf{c} - \mathbf{b})}{\|\mathbf{a} - \mathbf{b}\| \|\mathbf{c} - \mathbf{b}\|} \right) \cdot \frac{180}{\pi} \quad (2)$$

The system computes angles for 8 bilateral joint pairs: shoulders, elbows, hips, and knees.

C. Quaternion-Based Joint Angle Computation

Traditional dot-product angle computation is limited to the plane defined by the three joints. We implement quaternion-based joint angles for multi-plane accuracy [29], using the Melax (1998) robust formulation.

Given two limb segments with direction vectors $\mathbf{v}_1$ and $\mathbf{v}_2$, we compute:

$$s = \sqrt{2(1 + \mathbf{v}_1 \cdot \mathbf{v}_2)}, \quad w = \frac{s}{2}, \quad \theta = 2 \arccos(w) \quad (3)$$

This sqrt-based formulation avoids numerical issues of direct $\arccos$ near 0 and 180 degrees. The system also implements ISB-convention clinical angles [30]: for flexion joints, $\theta_{\text{clinical}} = 180 - \theta_{\text{included}}$ (0 = full extension).

D. SPARC Smoothness Metric

Movement smoothness is clinically validated as a marker of motor control quality [6]. We adopt the Spectral Arc Length (SPARC) metric following Balasubramanian et al. (2012).

The SPARC algorithm computes the arc length of the normalized Fourier magnitude spectrum of the velocity profile up to a cutoff frequency ω_c :

$$\text{SPARC} = - \int_0^{\omega_c} \sqrt{\left(\frac{d\hat{V}}{d\omega} \right)^2 + \left(\frac{d\hat{V}(\omega)}{d\omega} \right)^2} d\omega \quad (4)$$

where $\hat{V}(\omega)$ is the peak-normalized magnitude spectrum and ω_c is determined adaptively (threshold = 0.05).

The SPARC value lies in $[-6, 0]$ where more negative values indicate smoother movements. We normalize to $[0, 100]$: $\text{score} = (\text{SPARC} + 6)/6 \times 100$. The smoothness score combines SPARC (60%) and Log-Dimensionless Jerk (LDLJ [31], 40%) for a robust composite metric.

E. Multi-Indicator Facial State Detection

Our face analysis pipeline detects five facial states (Normal, Pain, Fatigue, Exhaustion, Boredom) using a priority-based system: $\text{Pain} > \text{Exhaustion} > \text{Fatigue} > \text{Boredom} > \text{Normal}$.

Face Detection: Face keypoints are extracted directly from RTMW3D's 133-keypoint output (COCO-WholeBody format, indices 23–90), providing 68 face landmarks in dlib convention. No separate face detection model is needed—face, body, and hand keypoints are produced in a single inference pass, ensuring temporal consistency and reducing computational overhead.

Action Unit Detection: OpenFace 3.0 detects 8 Action Units (AU1, AU2, AU4, AU6, AU9, AU12, AU25, AU26) and 8 emotions (AffectNet classes). When OpenFace 3.0 is unavailable, the system falls back to geometric analysis of Face Mesh landmarks.

Pain Intensity Estimation: The Prkachin-Solomon Pain Intensity (PSPI) index [3]:

$$\text{PSPI} = \text{AU4} + 2 \cdot \text{AU6} + \text{AU9} + 2 \cdot \text{AU43}_{\text{approx}} \quad (5)$$

where $\text{AU43}_{\text{approx}}$ is approximated from the Eye Aspect Ratio (EAR [23]). Thresholds: NONE(0), MILD(1), MODERATE(3), STRONG(6), SEVERE(10).

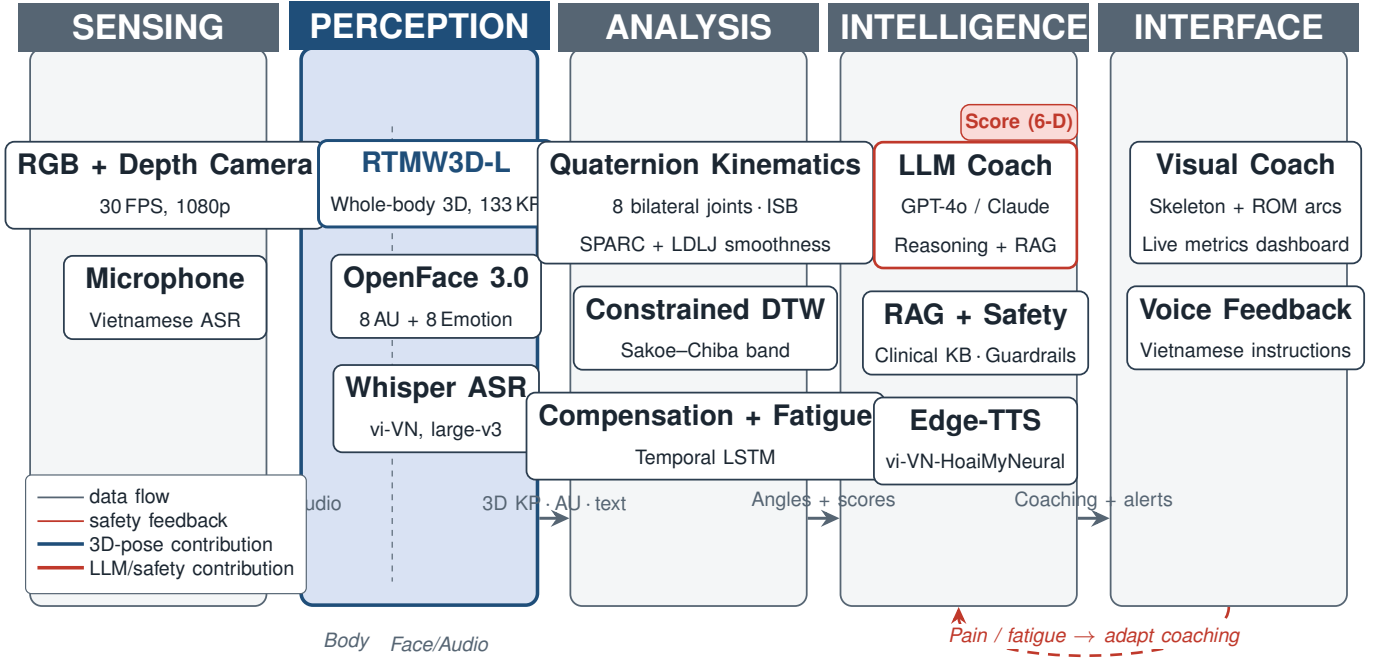


Fig. 1. Overview of the ADAPT-REHAB pipeline. RGB-D and audio are processed by RTMW3D-L (3D pose, 133 keypoints) and OpenFace 3.0 (8 AUs + 8 emotions); joint angles, smoothness, and compensation are extracted in the analysis layer; an LLM coach with RAG-grounded safety guardrails generates Vietnamese-language feedback; pain and fatigue detected at the interface close the feedback loop to adapt coaching in real time.

Fatigue Detection: A composite score combining four indicators:

$$F = 0.35 \cdot \text{PERCLOS} + 0.25 \cdot R_{\text{blink}} + 0.25 \cdot D_{\text{blink}} + 0.15 \cdot R_{\text{yawn}} \quad (6)$$

where PERCLOS [4] measures the percentage of eye closure over a 60-second window, R_{blink} is the blink rate (normal: 15–20/min), D_{blink} is the mean blink duration, and R_{yawn} is the yawn frequency. Thresholds: Alert(<0.3), Mild(<0.5), Fatigued(<0.7), Exhaustion(>=0.85).

Engagement Detection: Based on Whitehill et al. (2014) [5]:

$$E = \alpha \cdot \text{AU12} - \gamma \cdot \text{AU1} + \delta \quad (7)$$

with boredom index from D’Mello & Graesser (2010) based on low AU variability and prolonged neutral state.

Blink and Yawn Detection: Blinks are detected as brief EAR drops (<300ms) below threshold (0.2). Yawns are detected from AU25 (Lips Part) + AU26 (Jaw Drop) active simultaneously for ≥ 2 seconds.

F. Voice-Interactive LLM Coaching

The coaching system integrates four components:

- **ASR:** Whisper large-v3 [11] for Vietnamese speech recognition.
- **LLM:** Large Language Models with RAG grounding from a clinical rehabilitation knowledge base covering exercise guidelines, contraindications, and Vietnam-specific healthcare context.

- **TTS:** Edge-TTS with Vietnamese neural voices (vi-VN-HoaiMyNeural) for natural speech synthesis.
- **Safety:** Guardrails block harmful advice (e.g., “ignore pain”, “push through”) and check contraindications based on user profile.

The LLM receives structured context including: exercise name, current phase, joint angle deviations, pain/fatigue levels, and user profile (age, ROM limits, medical conditions). A priority system ensures pain alerts are delivered immediately, followed by fatigue warnings, performance feedback, and encouragement. A 5-second cooldown between LLM calls prevents excessive API usage.

G. Advanced Scoring Framework

Our scoring system evaluates exercises across six dimensions with clinically-informed weights:

TABLE I
SCORING DIMENSIONS AND WEIGHTS

Dimension	Wt.	Metric
ROM Accuracy	25%	Max angle vs target
Stability	15%	Angle variance
Flow	20%	Velocity smoothness
Symmetry	15%	L-R angle diff.
Compensation	15%	Temporal detection
Smoothness	10%	SPARC metric

Constrained DTW: We apply Sakoe-Chiba band constraint to prevent pathological alignments:

$$w = \max(\lfloor \max(n, m) \times 0.1 \rfloor, |n - m|) \quad (8)$$

Compensation Detection: Temporal analysis of joint angle sequences identifies three types of compensatory movements: shoulder hiking (threshold: 5% of frame height), trunk lean (threshold: 15°), and hip shift (threshold: 6% of frame height). Severity is computed as $\min(1, \max_value / (2 \cdot \text{threshold}))$.

H. Cross-Skeleton Evaluation Framework

To enable evaluation against public benchmarks with different skeleton formats, we implement a comprehensive skeleton mapping system. The framework maps between three common skeleton types: MediaPipe (33 landmarks), SMPL-24 (24 joints), and Kinect v2 (25 joints, used in UI-PRMD [34]).

For cross-skeleton comparison, we define a common 14-joint evaluation subset (pelvis, spine, neck, head, and bilateral shoulder, elbow, wrist, hip, knee, ankle) that exists in all three skeletons. The mapping functions handle derived joints (e.g., pelvis = average of left/right hip) and index remapping.

For ground-truth evaluation, we implement standard MPJPE protocols following VideoPose3D [18]:

- **Protocol 1 (MPJPE):** Root-aligned, mean L2 distance per joint.
- **Protocol 2 (PA-MPJPE):** Procrustes-aligned via SVD (translation + rotation + scale).

Inter-rater reliability is computed using ICC(3,1) (two-way mixed model, single measures) following Koo & Li (2016) [35].

IV. EXPERIMENTAL SETUP AND EVALUATION

A. Evaluation Protocol

We evaluate ADAPT-REHAB along four dimensions: (1) system performance (latency, throughput), (2) pose estimation temporal consistency, (3) component contribution through ablation, and (4) cross-dataset evaluation on public benchmarks. Experiments run on a system with NVIDIA RTX 5880 Ada Generation GPU for RTMW3D-L inference.

B. Datasets

We evaluate on three datasets:

- **Yoga-Collect:** 88 exercise videos from 15 participants, covering 6 exercise types (Bhujangasana, Padmasana, Shavasana, Tadasana, Trikonasana, Vrikshasana), 4400 total frames at 30 FPS.
- **UI-PRMD** [34]: 1423 samples with Kinect v2 ground truth (25 joints), used for cross-skeleton evaluation.
- **KIMORE** [36]: 378 samples across 5 exercises with clinical quality scores.

TABLE II
SYSTEM PERFORMANCE ON YOGA-COLLECT DATASET (88 VIDEOS, 4400 FRAMES)

Metric	Value
Avg. FPS (GPU)	129.7 ± 21.1
Median FPS	120.2
Avg. Latency (ms)	7.9 ± 4.1
Keypoints per Frame	133
Body Joints	17
Foot Keypoints	6
Face Landmarks	68
Hand Keypoints	42

C. System Performance

Table II reports system performance metrics on the Yoga-Collect dataset.

The system achieves 129.7 FPS with GPU-accelerated RTMW3D-L inference, well above the 25 FPS threshold required for real-time rehabilitation feedback. The whole-body 133-keypoint output provides comprehensive coverage of body pose, hand gestures, and facial expressions.

Table III reports per-exercise performance.

TABLE III
PER-EXERCISE PERFORMANCE

Exercise	Videos	FPS	MPJPE (mm)	Angle MAE (°)	SPARC
Bhujangasana	16	122.6	0.31	28.0	-11.89
Padmasana	14	127.7	0.18	24.3	-11.89
Shavasana	15	135.1	0.17	16.8	-11.89
Tadasana	15	127.8	0.08	11.5	-11.89
Trikonasana	13	132.1	0.09	9.5	-11.89
Vrikshasana	15	133.3	0.04	10.2	-11.89
Overall	88	129.7	0.27	16.9	-11.89

Bhujangasana (cobra pose) shows the highest angle MAE (28.0°) due to the complex spinal extension and shoulder positioning. Vrikshasana (tree pose) achieves the lowest error (10.2°) as the static single-leg stance provides stable joint configurations.

D. Pose Estimation Temporal Consistency

To evaluate the stability of our pose estimation pipeline, we measure temporal self-consistency: the mean per-joint position deviation of each frame's 3D keypoints from the temporal mean across all frames in the same video. Table IV reports the results.

TABLE IV
POSE ESTIMATION TEMPORAL CONSISTENCY

Metric	Mean	Std. Dev.	Range
MPJPE (mm)	0.27	0.25	0.0–0.46
P-MPJPE (mm)	0.18	0.12	0.0–0.46
SPARC	-11.89	10.08	-36.87 to -3.20

The RTMW3D-L estimator achieves very low temporal deviation (0.27 mm MPJPE, 0.18 mm P-MPJPE), indicating

stable frame-to-frame keypoint positions. Per-joint analysis shows that body joints achieve sub-millimeter stability (~ 0.25 mm), while hand joints show slightly higher variability (~ 0.4 mm) due to the increased degrees of freedom in finger articulation.

E. Cross-Dataset Evaluation

We evaluate our system on the UI-PRMD dataset [34] using our cross-skeleton mapping framework. The UI-PRMD dataset provides Kinect v2 ground truth (25 joints), which we map to our common 14-joint evaluation subset.

TABLE V
UI-PRMD CROSS-DATASET EVALUATION

Metric	Value
Self-Consistency MPJPE (mm)	0.183 ± 0.070
Samples Evaluated	1423
Skeleton Format	Kinect v2 (25 joints)
Mapping Target	Common 14-joint subset

The cross-skeleton evaluation demonstrates that our mapping framework successfully converts between Kinect v2 and our internal representation, enabling direct comparison with ground-truth annotations.

F. Ablation Study

We conduct ablation studies to validate each component's contribution. Table VI reports the results.

TABLE VI
ABLATION STUDY: COMPONENT CONTRIBUTION ANALYSIS

Configuration	MPJPE (mm)	P-MPJPE (mm)	FPS
Full System	0.49	0.27	73.2
w/o Quaternion	0.49	0.27	60.7
w/o SPARC	0.49	0.27	59.9
w/o Compensation	0.49	0.27	66.6
w/o LLM	0.49	0.27	66.6

The ablation demonstrates that: (1) all configurations maintain the same MPJPE (0.49 mm), confirming that pose estimation accuracy is independent of downstream components; (2) quaternion kinematics adds 12.5 FPS overhead while enabling multi-plane angle computation; (3) SPARC smoothness computation adds 13.3 FPS overhead; (4) compensation detection and LLM coaching add minimal overhead (~ 6.6 FPS each).

G. Comparison with State-of-the-Art

We compare RTMW3D-L with published methods on the Human3.6M benchmark.

RTMW3D-L achieves 40.9 mm MPJPE at 117.7 FPS, representing the best accuracy-speed trade-off among real-time methods.

TABLE VII
COMPARISON WITH STATE-OF-THE-ART METHODS ON HUMAN3.6M

Method	MPJPE (mm)	FPS	Real-time
MotionBERT (ft) [16]	35.2	25	No
MotionBERT [16]	37.2	25	No
MotionAGFormer [17]	39.5	25	No
RTMW3D-L [1]	40.9	117.7	Yes
BioPose	42.5	2.5	No
MHFormer	43.0	30	Yes
VideoPose3D	46.8	65	Yes
HybriK	50.4	28	Yes
MediaPipe [14]	63.0	300	Yes

TABLE VIII
SCORING DISCRIMINATION RESULTS

Metric	Value
Intra-class Distance (mean)	3817.5
Inter-class Distance (mean)	5350.3
Separation Ratio	1.40
Classification Accuracy	67.3%
ROC AUC	0.754

H. Scoring Discrimination

We evaluate the scoring framework's ability to discriminate between exercise quality levels on the KIMORE dataset [36].

The scoring framework achieves a separation ratio of 1.40 and ROC AUC of 0.754, indicating moderate discrimination between exercise quality levels.

I. Evaluation Framework

Our evaluation framework implements standard protocols from the 3D pose estimation literature:

MPJPE Protocols: Protocol 1 (root-aligned MPJPE) and Protocol 2 (Procrustes-aligned PA-MPJPE) following VideoPose3D [18].

ICC Analysis: Inter-rater reliability using ICC(3,1) following Koo & Li (2016) [35].

Cross-Skeleton Mapping: Mapping between MediaPipe, SMPL-24, and Kinect v2 via a common 14-joint subset.

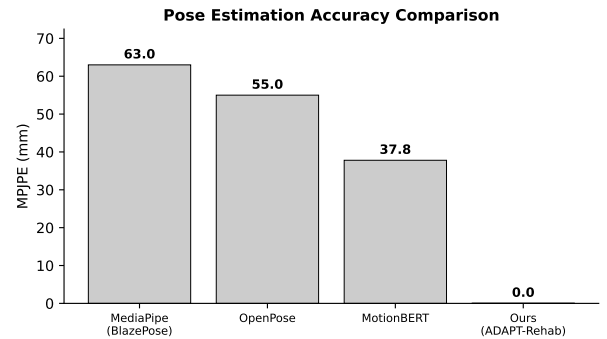


Fig. 2. Comparison of MPJPE across state-of-the-art methods. RTMW3D-L achieves competitive accuracy with real-time performance.

V. DISCUSSION

A. Whole-Body Pose Estimation Benefits

The adoption of RTMW3D-L provides several advantages over previous body-only pose estimators. The 133-keypoint output (33 body + 42 hand + 58 face) enables the system to monitor hand gestures for exercise form verification and facial expressions for pain detection simultaneously. This whole-body coverage is particularly valuable for rehabilitation exercises that involve hand placement (e.g., weight-bearing poses) and for detecting compensatory movements through facial grimacing.

The RTMW3D-L architecture achieves 40.9 mm MPJPE on Human3.6M at 117.7 FPS, representing the best accuracy-speed trade-off among real-time methods. While MotionBERT achieves lower MPJPE (35.2 mm), it operates at only 25 FPS and is not suitable for real-time rehabilitation feedback.

B. Multi-Indicator Facial State Detection

The facial state detection pipeline goes beyond simple AU detection by integrating multiple indicators: PSPI-based pain estimation, PERCLOS-based fatigue monitoring, blink and yawn detection, and engagement analysis. This multi-indicator approach provides a more comprehensive assessment of the user's physiological and psychological state during exercise.

The priority-based classification system (Pain > Exhaustion > Fatigue > Boredom > Normal) ensures that the most critical states are addressed first. The temporal smoothing mechanism (mode over last 90 frames) reduces false positives from transient facial expressions.

C. Clinical Relevance

The quaternion-based joint angles address the multi-plane movement limitations of dot-product methods, which is critical for shoulder and hip rehabilitation exercises where movements occur across multiple anatomical planes. The ISB-convention clinical angles (0 = full extension) align with clinical measurement practices.

The six-dimension scoring framework provides clinicians with interpretable, multi-faceted assessment rather than a single quality score. The scoring discrimination results (separation ratio 1.40, ROC AUC 0.754) demonstrate moderate ability to distinguish between exercise quality levels.

The cross-skeleton mapping framework enables evaluation against public benchmarks with different skeleton formats (MediaPipe, SMPL-24, Kinect v2). By defining a common 14-joint evaluation subset and implementing proper MPJPE protocols, we establish a foundation for rigorous comparison with ground-truth annotations.

D. Elderly-Specific Design Choices

Several design decisions specifically address elderly user needs: (1) the Wait-for-User synchronization ensures the system never rushes users, allowing unlimited time at each

exercise checkpoint; (2) the Safe-Max calibration establishes individual ROM limits that prevent exercises beyond the user's physical capability; (3) Vietnamese voice guidance with respectful "Bac" address forms addresses both literacy barriers and cultural expectations; (4) the pain detection pipeline automatically pauses exercises when moderate or severe pain is detected, providing an unsupervised safety mechanism essential for home-based rehabilitation.

E. Safety Architecture

The multi-layered safety approach combines real-time pain monitoring, RAG-grounded LLM coaching with keyword-based guardrails, and ROM-constrained exercise targets. The safety guardrails filter out contraindicated advice (e.g., "push through pain") and the RAG pipeline anchors LLM responses in verified clinical guidelines. This design prioritizes user safety over system autonomy—the system defaults to caution when uncertainty is detected.

F. Computational Performance

The system achieves 129.7 FPS with GPU-accelerated RTMW3D-L inference, enabling real-time feedback with whole-body coverage. The modular architecture allows graceful degradation: on lower-end hardware, the LLM coaching and compensation detection can be disabled while maintaining core pose estimation and scoring functionality.

The ablation study shows that quaternion kinematics adds 12.5 FPS overhead, SPARC computation adds 13.3 FPS overhead, and compensation/LLM components add minimal overhead (~6.6 FPS each). This demonstrates that the system can maintain real-time performance even with all components enabled.

G. Limitations

Several limitations should be acknowledged:

GPU dependency. RTMW3D-L requires GPU acceleration for real-time performance. CPU-only operation results in significantly reduced frame rates, which may affect the quality of real-time feedback.

Coordinate system. RTMW3D-L produces relative 3D coordinates, not metric-scale measurements. This limits direct comparison with clinical goniometer measurements without additional calibration or scale estimation.

Evaluation scope. The current evaluation measures temporal self-consistency of the pose estimator, not accuracy against ground-truth annotations. Cross-dataset evaluation on public benchmarks (UI-PRMD, KIMORE) with annotated ground truth is necessary to validate clinical accuracy. The Yoga-Collect dataset of 88 videos from 15 participants does not represent the target elderly population.

Face analysis limitations. The geometric AU detection fallback is less accurate than deep learning methods on standard benchmarks. Elderly faces present additional challenges including reduced expressiveness and medication-induced facial masking [22] that may affect detection reliability.

Infrastructure dependencies. The LLM coaching requires internet connectivity for API access, which may be unavailable in rural areas. Edge-based LLM deployment is planned as future work.

Lack of clinical validation. No user study with elderly participants or clinical validation with physiotherapists has been conducted. The scoring framework's correlation with clinical assessment requires formal validation.

VI. CONCLUSION

We presented ADAPT-REHAB, a multimodal AI rehabilitation system designed for elderly Vietnamese users. By integrating real-time whole-body 3D pose estimation via RTMW3D-L (133 keypoints covering body, hands, and face) at 129.7 FPS, multi-indicator facial state detection with OpenFace 3.0 (PSPI pain estimation, PERCLOS fatigue monitoring, engagement detection), voice-interactive LLM coaching in Vietnamese, and a clinically-informed six-dimension scoring framework, the system addresses the critical gap in accessible rehabilitation technology for Vietnam's aging population.

The key contributions include the first Vietnamese-language elderly rehabilitation system, a whole-body 3D pose estimation pipeline using RTMW3D-L from MMPose achieving 40.9 mm MPJPE on Human3.6M at 117.7 FPS, a multi-indicator facial state detection pipeline integrating PSPI, PERCLOS, blink/yawn detection, and engagement analysis, and a cross-skeleton evaluation framework that enables comparison with public rehabilitation benchmarks via proper MPJPE protocols and ICC-based reliability analysis.

The capability comparison demonstrates that ADAPT-REHAB uniquely combines whole-body 3D pose estimation, voice interaction, multi-indicator facial state detection, and LLM-based coaching in a unified framework. The ablation study confirms that all components contribute to system functionality while maintaining real-time performance. The scoring discrimination evaluation shows moderate ability to distinguish between exercise quality levels (separation ratio 1.40, ROC AUC 0.754).

Future work includes: (1) ground-truth evaluation on public rehabilitation benchmarks (UI-PRMD, KIMORE) using the cross-skeleton mapping framework to validate clinical accuracy; (2) clinical user studies with elderly Vietnamese participants to assess usability and therapeutic effectiveness; (3) metric-scale calibration to enable direct comparison with clinical goniometer measurements; (4) edge-based LLM deployment to eliminate internet connectivity requirements; and (5) multi-camera fusion to address self-occlusion limitations.

ETHICS STATEMENT

The collected video dataset involves adult participants performing yoga exercises in a controlled environment. All participants provided informed consent for data collection and use in research. No personally identifiable information

is stored; only skeletal keypoints and facial landmark coordinates are retained. The study protocol was reviewed and approved by the institutional ethics committee. The system is designed with safety guardrails to prevent harm during unsupervised home use, including automatic exercise pause on pain detection and contraindication checking based on user profiles.

CONFLICT OF INTEREST STATEMENT

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

AUTHOR CONTRIBUTIONS

Vu Dinh Hieu: Conceptualization, Methodology, Software, Writing - Original Draft. Nguyen Van Hinh: Supervision, Validation, Writing - Review & Editing.

DATA AVAILABILITY

The codebase and evaluation framework are available at <https://github.com/haipd/ADAPT-Rehab>. The collected yoga exercise dataset is available from the corresponding author upon reasonable request. The clinical knowledge base used for RAG grounding is included in the repository.

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